Combine GTSeek

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Contents

Combining datasets for recapture analysis: LGC and GTSeek

Some of the samples with DNA yields too low for capture-seq at Rapid Genomics were sent to GTSeek for genotyping-in-thousands by sequencing (GT-seq) genotyping (Campbell et al. 2015) using a panel of markers developed by Dana Morin at Mississippi State University. We want to combine the two datasets for recapture analyses that follow. To do so, there are several things we need to do here:

1. compile metadata for the samples and add this to our `sampleMetadata` file
2. extract SNPs targeted by the GT-seq panel for hair samples genotyped by capture-seq
3. apply some modest locus-level filters (quality, genotype depth) to LGC data
4. apply some individual-level filters to GTSeek data (based on the individual fuzziness index (IFI) - a measure of potential contamination)
5. combine the two datasets
6. record the number of SNPs genotyped for all samples in our `sampleMetadata` file
7. drop samples that are not sufficiently genotyped prior to recapture analyses
    - for GTSeek samples, at least 100 SNPs genotyped
    - for LGC samples, <50% missing data (in the combined dataset)

We'll walk through these steps below. At the end of the markdown, we include a short section evaluating the power of the marker set via calculation of probabilities of identity for random individuals (p_{ID}) and for siblings (p_{IDsib}). These measure the chance that two unrelated individuals, or a pair of siblings, would share identical genotypes.

Compile metadata

First, let's load the data and some files with information about the samples (which were submitted to GTSeek in four batches)...

```
gt <- read.csv("../input/UGA_bbh_Prod1_all_Genotypes.csv", header = TRUE)
gt[1:5,1:8]
```

```
##                                Sample Raw.Reads On.Target.Reads X.On.Target
## 1   GTseek_024_384i5_001_P098.001_5-23    659842          116896       17.72
## 2 GTseek_024_384i5_002_P098.003_928-23    918083          137368       14.96
## 3  GTseek_024_384i5_003_P098.001_65-23    763212            3067        0.40
## 4 GTseek_024_384i5_004_P098.003_929-23    556873          425201       76.36
## 5  GTseek_024_384i5_005_P098.001_82-23    876647            3948        0.45
##    X.GT  IFI NW_025576306.1_4455299 NW_025576307.1_2027013
## 1 81.63 0.63                   GG                    00
## 2 93.20 2.06                   GA                    CC
## 3 61.22 5.22                   GG                    00
## 4 76.19 6.42                   00                    TC
## 5 50.34 8.07                   00                    TC
```

```
#Create a ugaID column with sample identifiers
gt$ugaID <- sapply(gt$Sample, FUN=function(x) strsplit(x, split="_")[[1]][6])

#Create an object for metadata
gt.smd <- data.frame(gtsID = gt$Sample,
                     ugaID = gt$ugaID,
                     is.SCR = NA,
                     is.rep = NA,
                     week=NA,
                     year=NA,
                     snare=NA,
                     utmSnare=NA,
                     fqfile=NA,
                     knownSEX=NA,
                     sampSOURCE=NA,
                     region=NA,
                     nSNPgt=NA)
#To fill in the utm snares, we'll need the translatable.csv file
# this was generated in `1_snareLocations`
translatable <- read.csv("../sampleinfo/translatable.csv", header = TRUE)

#We also need the source of metadata for these GTSeek samples
#Read sample metadata into R
tmp1 <- read.csv("../input/GTSeek_rd1_Metadata.csv", header = TRUE)
tmp2 <- read.csv("../input/GTSeek_rd2_Metadata.csv", header = TRUE)
tmp3 <- read.csv("../input/GTSeek_rd3_Metadata.csv", header = TRUE)
```

```
tmp4 <- read.csv("../input/GTSeek_rd4_Metadata.csv", header = TRUE)
#Some additional metadata for samples that were originally submitted to LGC
lgcinfo <- read.csv("../input/Finalized Hair Samples w Metadata_Carr.csv",
                    header = TRUE)


colnames(tmp4)[which(colnames(tmp4) == "Site.ID")] <- "Site.Name"
tmp4 <- tmp4[,-which(colnames(tmp4) == "RandomID")]
colnames(tmp4)[which(colnames(tmp4) == "Hair.ID")] <- "RandomID"

keepcols <- c("RandomID","Site.Name","Week","Year")
sampinfo <- rbind(tmp1[,keepcols],
                  tmp2[,keepcols],
                  tmp3[,keepcols],
                  tmp4[,keepcols])

#Clean the dataset
#A few samples had incorrect or missing names, or were duplicated unintentionally
#624-24 was duplicated
gt <- gt[-which(gt$ugaID == "624-24")[1],] #Drop it from gt
gt.smd <- gt.smd[-which(gt.smd$ugaID=="624-24")[1],] # from gt.smd
sampinfo <- sampinfo[-which(sampinfo$RandomID == "624-24")[1],] # and from sampinfo

#A couple of mislabeled samples
gt.smd$ugaID[which(gt.smd$ugaID == "DC-99427AB")] <- "DC-99427B"
gt.smd$ugaID[which(gt.smd$ugaID == "DC-99427Sow")] <- "DC-87805Sow"

#A sample with missing name
gt.smd$ugaID[which(gt.smd$ugaID == "#REF!")] <- "UNK"

#One hair snare without a leading zero in the name
sampinfo$Site.Name[which(sampinfo$Site.Name == "HS 44")] <- "HS 044"

#Going through the loop to fill in metadata...
for(i in 1:nrow(gt.smd)) {
  #gtsID & ugaID are already filled in
  if(gt.smd$ugaID[i] %in% lgcinfo$RandomID) {
    is.LGC <- TRUE
  } else {
    is.LGC <- FALSE
  }

  #Account for NTC samples
  if(length(grep("NTC",gt.smd$ugaID[i])) == 1) {
    next
  }

  #Fill in is.rep
  if(length(grep("R$", gt.smd$ugaID[i])) == 1) {
    gt.smd$is.rep[i] <- ifelse(
      length(which(gt.smd$ugaID == sub("R$","",gt.smd$ugaID[i]))) == 0,
      FALSE, TRUE)
    tmp.ID <- sub("R$","",gt.smd$ugaID[i])
```

```
} else {
  gt.smd$is.rep[i] <- ifelse(
    length(which(gt.smd$ugaID == paste0(gt.smd$ugaID[i],"R"))) == 0,
    FALSE, TRUE)
  tmp.ID <- gt.smd$ugaID[i]
}

#Account for a few samples that are not in sampinfo
if(gt.smd$is.rep[i] == FALSE &
   length(which(sampinfo$RandomID == tmp.ID)) == 0) {
  gt.smd$is.SCR[i] <- FALSE
  next
}

gt.smd$week[i] <- sampinfo$Week[sampinfo$RandomID == tmp.ID]
gt.smd$year[i] <- sampinfo$Year[sampinfo$RandomID == tmp.ID]

if(length(grep("HS",sampinfo$Site.Name[sampinfo$RandomID == tmp.ID])) > 0) {
  gt.smd$snare[i] <- sampinfo$Site.Name[sampinfo$RandomID == tmp.ID]
  if(gt.smd$year[i] == 2023) {
    gt.smd$utmSnare[i] <- translatable$UTMname[which(translatable$name23 == gt.smd$snare[i])]
  } else if(gt.smd$year[i] == 2024) {
    gt.smd$utmSnare[i] <- translatable$UTMname[which(translatable$name24 == gt.smd$snare[i])]
  }
  gt.smd$region[i] <- "CGA"
  gt.smd$sampSOURCE[i] <- "Hair Sample"
  gt.smd$is.SCR[i] <- TRUE
} else {
  gt.smd$snare[i] <- NA
  gt.smd$utmSnare[i] <- NA
  gt.smd$is.SCR[i] <- FALSE
  gt.smd$sampSOURCE[i] <- sampinfo$Site.Name[sampinfo$RandomID == tmp.ID]
  if(gt.smd$year[i] == 2012) {
    gt.smd$region[i] <- "CGA-H"
  } else {
    gt.smd$region[i] <- "CGA"
  }
}

#nSNPgt is NA for now...
gt.smd$nSNPgt[i] <- NA
#Some of the sexes are probably known, but not recorded in our metadata...
gt.smd$knownSEX[i] <- NA

#If sample was submitted to LGC, pull additional metadata
if(is.LGC) {
  which.row <- which(lgcinfo$Hair.ID == tmp.ID)
  gt.smd$sampSOURCE[i] <- lgcinfo$What[which.row]
  gt.smd$knownSEX[i] <- lgcinfo$Sex[which.row]
  gt.smd$region[i] <- lgcinfo$Region[which.row]
  rm(which.row)
}
rm(is.LGC, tmp.ID)
```

```
}

gt.smd[1:10,]
```

```
##                                  gtsID  ugaID is.SCR is.rep week year  snare
## 1    GTseek_024_384i5_001_P098.001_5-23   5-23   TRUE  FALSE    6 2023 HS 012
## 2  GTseek_024_384i5_002_P098.003_928-23 928-23   TRUE  FALSE    4 2023 HS 071
## 3   GTseek_024_384i5_003_P098.001_65-23  65-23   TRUE  FALSE    7 2023 HS 009
## 4  GTseek_024_384i5_004_P098.003_929-23 929-23   TRUE  FALSE    5 2023 HS 025
## 5   GTseek_024_384i5_005_P098.001_82-23  82-23   TRUE  FALSE    5 2023 HS 005
## 6  GTseek_024_384i5_006_P098.003_930-23 930-23   TRUE   TRUE    7 2023 HS 028
## 7  GTseek_024_384i5_007_P098.001_174-23 174-23   TRUE  FALSE    3 2023 HS 072
## 8  GTseek_024_384i5_008_P098.003_931-23 931-23   TRUE  FALSE    7 2023 HS 032
## 9  GTseek_024_384i5_009_P098.001_258-23 258-23   TRUE  FALSE    3 2023 HS 072
## 10 GTseek_024_384i5_010_P098.003_932-23 932-23   TRUE  FALSE    4 2023 HS 116
##          utmSnare fqfile knownSEX  sampSOURCE region nSNPgt
## 1  258412x3586648     NA     <NA> Hair Sample    CGA     NA
## 2  266900x3593042     NA     <NA> Hair Sample    CGA     NA
## 3  259209x3585507     NA     <NA> Hair Sample    CGA     NA
## 4  265276x3591787     NA     <NA> Hair Sample    CGA     NA
## 5  252875x3588670     NA     <NA> Hair Sample    CGA     NA
## 6  258399x3594245     NA     <NA> Hair Sample    CGA     NA
## 7  266197x3594132     NA     <NA> Hair Sample    CGA     NA
## 8  263672x3593292     NA     <NA> Hair Sample    CGA     NA
## 9  266197x3594132     NA     <NA> Hair Sample    CGA     NA
## 10 269531x3605091     NA     <NA> Hair Sample    CGA     NA
```

Extract SNPs

Now we'll use the column names of the `gt` object to create a list of SNPs to extract from our raw vcf file (from LGC) using the `--positions` argument to `vcftools`. In other words, add the loci genotyped by GTSeek to the set of markers that we're using for individual identification. Note that some of these SNPs will not have a minor allele frequency below the filter applied in `3_filterSCR`. We will retain them nonetheless to avoid too much missing data for samples processed via GTSeek.

```
#Column names, avoiding info columns, sexing loci columns, and the ugaID column
gtsSNPs <- colnames(gt)[-c(1:6,grep("Uam",colnames(gt)),ncol(gt))]
length(gtsSNPs)
```

```
## [1] 141
```

```
CPOS <- data.frame(CHROM=rep(NA, length(gtsSNPs)), POS=rep(NA, length(gtsSNPs)))

#column names are CHROM_POS - CHROMs are NW_#########.1
# e.g., NW_025576306.1_4455299
CPOS$CHROM <- sapply(gtsSNPs, FUN=function(x)
  paste(strsplit(x, split="_")[[1]][1:2], collapse="_"))
CPOS$POS <- sapply(gtsSNPs, FUN=function(x)
  strsplit(x, split="_")[[1]][3])

head(CPOS)
```

```
##            CHROM      POS
## 1 NW_025576306.1  4455299
## 2 NW_025576307.1  2027013
## 3 NW_025576331.1  1584277
## 4 NW_025576331.1   441039
## 5 NW_025576393.1  1735059
## 6 NW_025576453.1 20968993
```

```
#Check for loci that are ALREADY in LGC_SCR.vcf
filtSNPs <- data.frame(CHROM=system("grep -v ^# LGC_SCR.vcf | cut -f 1", intern=TRUE),
                       POS = system("grep -v ^# LGC_SCR.vcf | cut -f 2", intern=TRUE))

#Avoid including these loci!!
present <- c()
for(i in 1:nrow(CPOS)) {
  tmp.ind <- which(filtSNPs$CHROM == CPOS$CHROM[i] &
                     filtSNPs$POS == CPOS$POS[i])
  if(length(tmp.ind) > 0) {
    present <- c(present, i)
  }
}
CPOS <- CPOS[-present,]
length(present)
```

```
## [1] 34
```

```
write.table(CPOS, file = "GTSeek_SNPs.txt",
            quote = FALSE, row.names = FALSE, col.names = FALSE)
```

Now extract data for these markers using `vcftools`...

```
vcftools --vcf UGA_173101_SNPs_Raw.vcf --out GTSeek_SNPs \
          --positions GTSeek_SNPs.txt --recode

#How many are in the output?
grep -v ^# GTSeek_SNPs.recode.vcf | wc -l
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf UGA_173101_SNPs_Raw.vcf
    --out GTSeek_SNPs
    --positions GTSeek_SNPs.txt
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 99 out of a possible 319680 Sites
Run Time = 9.00 seconds
99
```

Filter LGC data

Now apply some modest filters (quality and genotype depth only) to this small dataset to avoid including bad genotypes or markers...

```
#First min quality
vcftools --vcf GTSeek_SNPs.recode.vcf --out LGC_GTS1 \
         --minQ 30 --recode

#Then min genotype depth
vcftools --vcf LGC_GTS1.recode.vcf --out LGC_GTS2 \
         --minDP 10 --recode

#Remove any half-calls that are included in this file
grep ^# LGC_GTS2.recode.vcf > preLGC_GTS.vcf
grep -v ^# LGC_GTS2.recode.vcf | sed -e 's/\.\/[0,1]/\.\/\./g' | sed -e 's/[0,1]\/\./\.\/\./g' >> preLG

#Then convert to genotype only format with plink2
plink2 --vcf preLGC_GTS.vcf --out LGC_GTS --recode vcf-4.2 --silent

#Combine these SNPs with those from your previous filtering effort
# using 'bcftools concat' - need to sort and index the file first
bcftools sort LGC_SCR.vcf -O b -o LGC_SCR.bcf
bcftools sort LGC_GTS.vcf -O b -o LGC_GTS.bcf
bcftools index LGC_SCR.bcf
bcftools index LGC_GTS.bcf
bcftools concat -a LGC_SCR.bcf LGC_GTS.bcf -O v -o LGC_allSCRsnps.vcf

#Cleanup
rm *LGC_GTS* GTSeek_SNPs.recode.vcf
```

```
VCFtools - 0.1.17
```

```
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf GTSeek_SNPs.recode.vcf
    --minQ 30
    --out LGC_GTS1
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
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Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 99 out of a possible 99 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf LGC_GTS1.recode.vcf
    --minDP 10
    --out LGC_GTS2
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
```

```
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 99 out of a possible 99 Sites
Run Time = 0.00 seconds
Writing to /tmp/bcftools.09BVO4
Merging 1 temporary files
Done
Cleaning
Writing to /tmp/bcftools.Yem6kP
Merging 1 temporary files
Done
Cleaning
Checking the headers and starting positions of 2 files
```

Filter GTSeek data

The GTSeek dataset has some samples that are poorly genotyped and some that may be contaminated. With that in mind, we're going to drop some samples before combining them with the rest of the data. Before we get to those filters, let's look at the data a little more closely.

GT-seq Performance

First just the total number of reads per sample in the dataset...

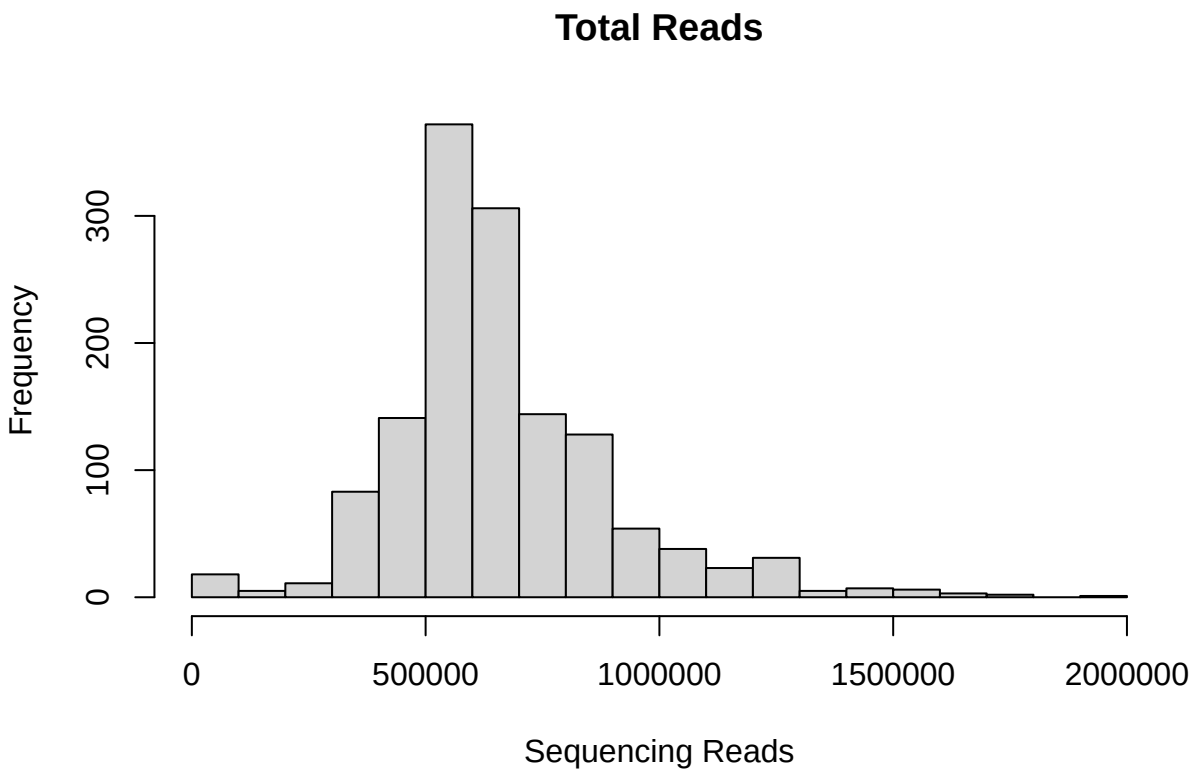
```
#Total Reads
quantile(gt$Raw.Reads) #There are some outliers
```

```
##        0%       25%       50%       75%      100%
##     539.0  531872.5  614148.5  776256.2 1974039.0
```

```
mean(gt$Raw.Reads)
```

```
## [1] 660728.8
```

```
hist(gt$Raw.Reads, breaks = 20,
     main = "Total Reads", xlab = "Sequencing Reads")
```



Now the percentage of loci genotyped per sample...

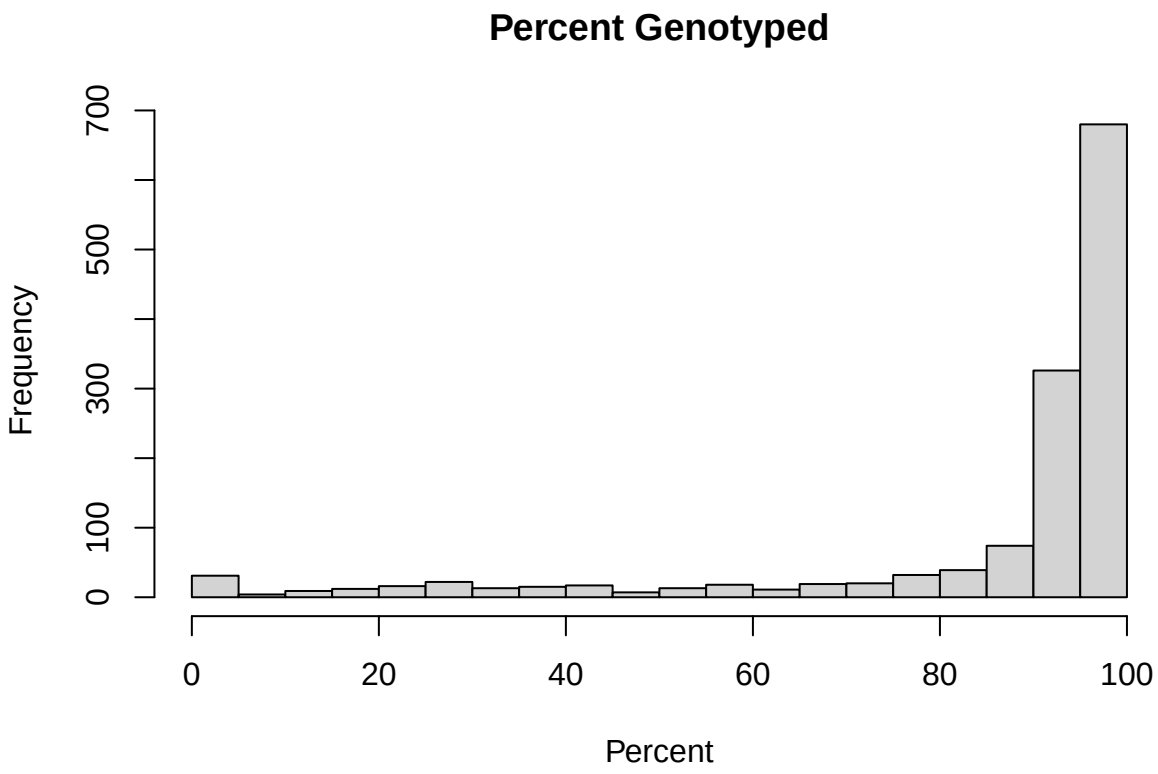
```
#Percent genotyped
quantile(gt$X.GT)
```

```
##    0%   25%   50%   75%  100%
##  0.00 88.44 94.56 95.92 98.64
```

```
mean(gt$X.GT)
```

```
## [1] 84.5604
```

```
hist(gt$X.GT, breaks = 20,
     main = "Percent Genotyped", xlab = "Percent")
```




```
length(which(gt$X.GT > 90))/nrow(gt) #What proportion are genotyped at >90%
```

```
## [1] 0.7300435
```

Finally, the *individual fuzziness index* (IFI). This index is described briefly in the header for some of the functions in Nate Campebell's GTseek_utils GitHub repository (the "GTseq_Genotyper" function, for instance). In the header, IFI is described as:

"*a measure of DNA cross contamination... calculated using read counts from background signal at homozygous and No-Call loci. Low scores are better than high scores.*"

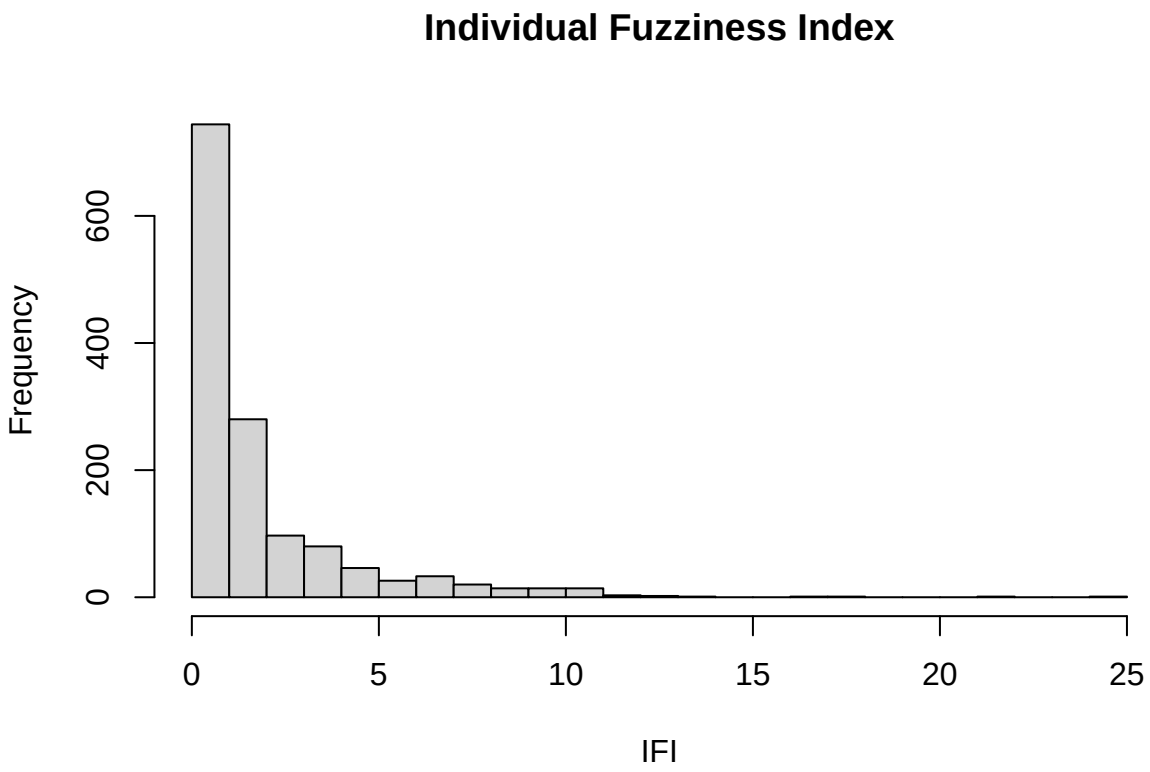
```
#IFI
quantile(gt$IFI)
```

```
##    0%   25%   50%   75%  100%
##  0.00  0.44  0.93  2.07 25.00
```

```
mean(gt$IFI)
```

```
## [1] 1.848534
```

```
hist(gt$IFI, breaks = 20,
     main = "Individual Fuzziness Index", xlab = "IFI")
```



Does IFI tell us anything about error rates? To assess this, we can look at our technical replicates...

```
#Find tech. reps
repID <- c(gt$ugaID[grep("-23R", gt$ugaID)],
           gt$ugaID[grep("-24R", gt$ugaID)])
ogID <- gsub("R","",repID)
comp.rep <- data.frame(ogID, repID,
                       og.XGT=NA, rep.XGT=NA,
                       ogIFI=NA, repIFI=NA,
                       overlap=NA, mismatch=NA, error=NA)
for(i in 1:nrow(comp.rep)) {
  #Check that both samples are present
  tmp.len <- length(which(gt$ugaID %in% c(repID[i], ogID[i])))
  if(tmp.len != 2) {
    stop(paste("Ooops! There are",tmp.len,"samples for",ogID[i]))
  }
  comp.rep$og.XGT[i] <- gt$X.GT[gt$ugaID == comp.rep$ogID[i]]
  comp.rep$rep.XGT[i] <- gt$X.GT[gt$ugaID == comp.rep$repID[i]]
  comp.rep$ogIFI[i] <- gt$IFI[gt$ugaID == comp.rep$ogID[i]]
  comp.rep$repIFI[i] <- gt$IFI[gt$ugaID == comp.rep$repID[i]]

  tmp.gt <- gt[which(gt$ugaID %in% c(comp.rep$ogID[i],
                                     comp.rep$repID[i])),
               -c(1:6,ncol(gt))]
  #Keep only overlapping calls
  tmp.gt <- tmp.gt[,which(apply(tmp.gt, 2,
```

```
                                FUN=function(x)
                                  sum(x %in% c("0","00")))==0)]
  comp.rep$overlap[i] <- ncol(tmp.gt)
  comp.rep$mismatch[i] <- sum(tmp.gt[1,] != tmp.gt[2,])
  comp.rep$error[i] <- comp.rep$mismatch[i] / comp.rep$overlap[i]
  rm(tmp.gt, tmp.len)
}

comp.rep
```

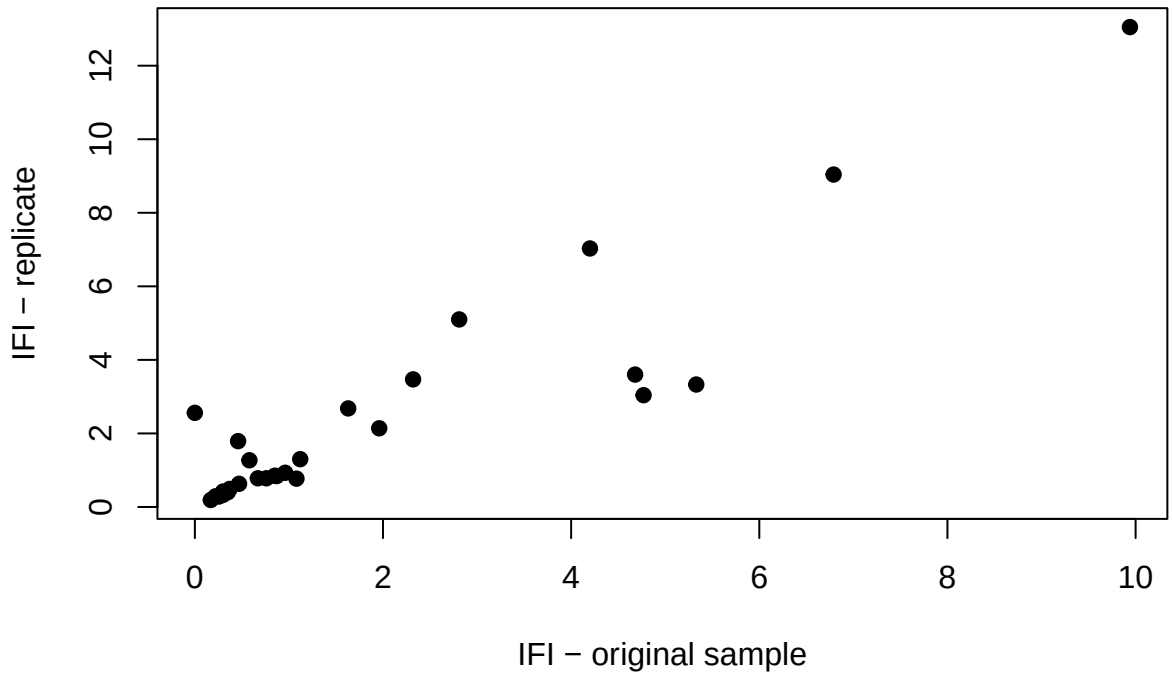
```
##       ogID    repID og.XGT rep.XGT ogIFI repIFI overlap mismatch       error
## 1   930-23  930-23R  95.92   96.60  1.12   1.30     141        0 0.000000000
## 2   781-23  781-23R  95.92   95.92  0.67   0.78     140        0 0.000000000
## 3   651-23  651-23R  94.56   95.92  0.47   0.63     139        0 0.000000000
## 4   798-23  798-23R  95.92   96.60  0.30   0.42     141        0 0.000000000
## 5   716-23  716-23R  83.67   87.76  4.68   3.60     112        7 0.062500000
## 6   935-23  935-23R  95.92   96.60  0.30   0.33     140        0 0.000000000
## 7   424-23  424-23R  69.39   36.05  5.33   3.33      39        5 0.128205128
## 8   466-23  466-23R  19.05   19.05  4.77   3.04       7        0 0.000000000
## 9   286-23  286-23R   3.40    2.04  0.00   2.56       2        0 0.000000000
## 10  311-23  311-23R  52.38   30.61  1.96   2.14      38        3 0.078947368
## 11  564-24  564-24R  94.56   95.24  0.85   0.85     139        0 0.000000000
## 12  517-24  517-24R  51.70   34.01  6.79   9.04      33        3 0.090909091
## 13  726-24  726-24R  84.35   75.51  4.20   7.03      98       13 0.132653061
## 14  980-24  980-24R  97.96   97.28  0.22   0.29     143        0 0.000000000
## 15 1306-24 1306-24R  96.60   96.60  0.25   0.29     141        0 0.000000000
## 16  567-24  567-24R  94.56   95.92  0.76   0.78     139        0 0.000000000
## 17  881-24  881-24R  95.92   92.52  0.46   1.79     134        0 0.000000000
## 18 2972-24 2972-24R  95.92   96.60  0.58   1.27     139        0 0.000000000
## 19 1384-24 1384-24R  73.47   66.67  9.94  13.05      78       29 0.371794872
## 20  933-24  933-24R  94.56   95.24  1.08   0.77     137        0 0.000000000
## 21 1522-24 1522-24R  96.60   95.92  0.96   0.93     141        1 0.007092199
## 22 3058-24 3058-24R  96.60   96.60  0.26   0.29     141        0 0.000000000
## 23 2665-24 2665-24R  94.56   95.24  0.87   0.84     138        0 0.000000000
## 24 2869-24 2869-24R  97.28   96.60  0.37   0.49     142        0 0.000000000
## 25 2926-24 2926-24R  91.84   87.76  2.32   3.47     122        0 0.000000000
## 26  151-24  151-24R  95.92   93.20  0.35   0.40     136        0 0.000000000
## 27  396-24  396-24R  89.12   84.35  1.63   2.68     118        1 0.008474576
## 28  191-24  191-24R  24.49   22.45  2.81   5.10      16        4 0.250000000
## 29  500-24  500-24R  95.92   96.60  0.17   0.19     141        0 0.000000000
```

```
cor.test(comp.rep$repIFI, comp.rep$ogIFI)
```

```
##
##  Pearson's product-moment correlation
##
## data:  comp.rep$repIFI and comp.rep$ogIFI
## t = 12.004, df = 27, p-value = 2.464e-12
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.8305545 0.9609892
## sample estimates:
```

```
##       cor
## 0.9177074
```

```
plot(comp.rep$repIFI ~ comp.rep$ogIFI,
     xlab = "IFI - original sample",
     ylab = "IFI - replicate",
     pch = 19)
```

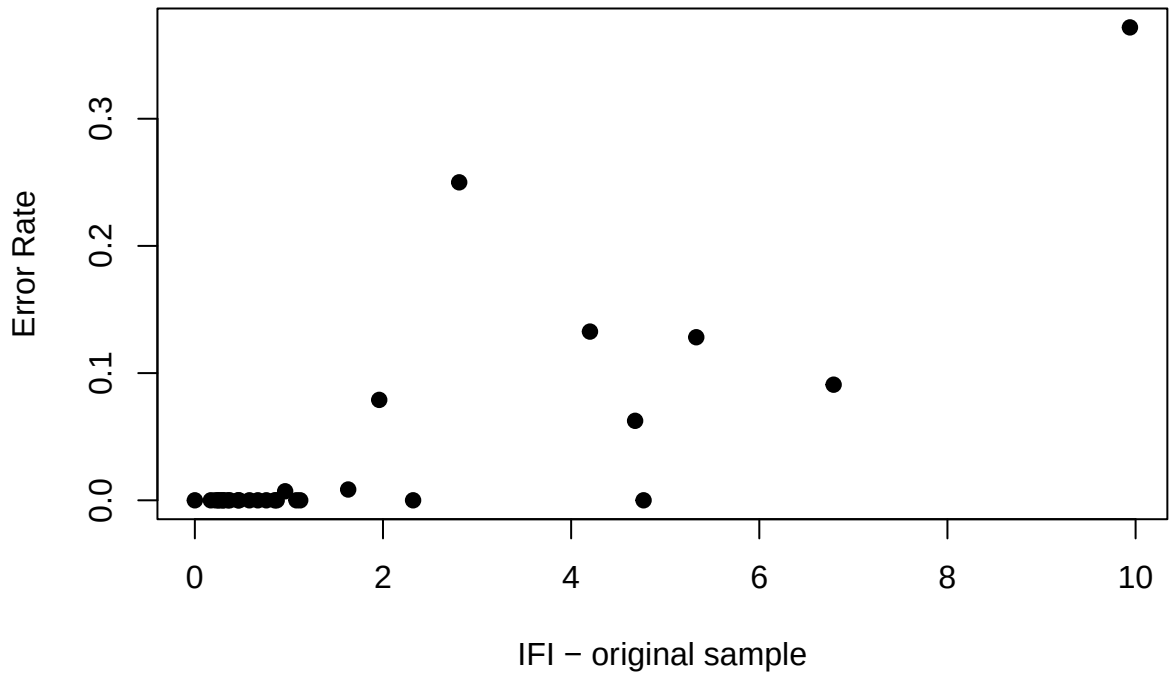



```
cor.test(comp.rep$ogIFI, comp.rep$error)
```

```
##
##  Pearson's product-moment correlation
##
## data:  comp.rep$ogIFI and comp.rep$error
## t = 6.7925, df = 27, p-value = 2.7e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.6033886 0.8990458
## sample estimates:
##       cor
## 0.7942497
```

```
plot(comp.rep$error ~ comp.rep$ogIFI,
     xlab = "IFI - original sample",
```

```
     ylab = "Error Rate",
     pch=19)
```



So the IFI is very similar between the two technical replicates, and it does seem to be predictive of mismatches between the replicates. This should be useful for identifying poorly genotyped samples that should be removed from the dataset.

Individual filtering

It seems reasonable to remove samples for either of two reasons, i) low rate of genotyping or ii) high IFI. In their paper describing GT-seq, Campbell et al. (2015) considered samples that were genotyped at less than 50% of SNPs as failed samples. We'll apply a more stringent filter for retaining samples below. For now, drop samples due to high IFI only...

```
#Identify samples with high IFI
drop.me <- which(gt$IFI > 2)
length(drop.me)
```

```
## [1] 354
```

```
#Now remove from genotype and metadata
nrow(gt)
```

```
## [1] 1378
```

```r
gt <- gt[-drop.me,]
nrow(gt)
```

```
## [1] 1024
```

Combine LGC and GTSeek data

Now convert genotypes to VCF format (`0/0`, `0/1`, `1/1`), taking care to assign `REF` alleles to `0` and `ALT` alleles to `1`, to match the conventions in the VCF. Then combine with `LGC_SCR.vcf` and check that `PLINK`, `vcftools`, and `vcfR` all recognize the combined output file as a vcf. If this works, you should be able to calculate the pairwise kinship coefficients in `PLINK2` and perform the clone analysis in `COLONY` to identify recaptures across the combined dataset.

```r
#First create a file with CHROM, POS, REF, and ALT for all SNPs in LGC_SCR
# This will be used to convert the CC, CT, TT genotypes to 0/0, 0/1, 1/1
# with appropriate coding for ref and alt alleles (0 an 1, respectively)
system("echo 'CHROM\tPOS\tREF\tALT' > scrSNPs.txt")
system("cat scrSNPs.txt", intern = TRUE)
```

```
## [1] "CHROM\tPOS\tREF\tALT"
```

```r
#SNP rows of the vcf | CHROM, POS, REF, and ALT columns
system("grep -v ^# LGC_allSCRsnps.vcf | cut -f 1-2,4-5 >> scrSNPs.txt")
system("wc -l scrSNPs.txt")
system("head scrSNPs.txt")

#Read this file with CHROM, POS, REF, and ALT for all SNPs into R
alleleState <- read.table("scrSNPs.txt", header = TRUE)
head(alleleState)
```

```
##            CHROM     POS REF ALT
## 1 NW_025576306.1 1250049   A   G
## 2 NW_025576306.1 1625051   C   A
## 3 NW_025576306.1 4425067   C   T
## 4 NW_025576306.1 4450032   A   G
## 5 NW_025576306.1 4455299   A   G
## 6 NW_025576306.1 5225027   T   C
```

```r
#Focus on genotypes at SNPs NOT used for sex determination
geno <- gt[,-c(1:6,grep("Uam", colnames(gt)),ncol(gt))]
dim(geno)
```

```
## [1] 1024  141
```

```r
#Make an object to fill in with vcf-formatted genotype data
vcfFormat <- matrix(data = "./.", nrow = nrow(alleleState), ncol = nrow(geno))
for(a in 1:nrow(vcfFormat)) {
  #Column names of the GTSeek data file are CHROM_POS
  tmp.colname <- paste0(alleleState$CHROM[a],"_",alleleState$POS[a])
```

```
  tmp.col <- which(colnames(geno) == tmp.colname)
  if(length(tmp.col) > 0) { #If this SNP from the LGC_SCR vcf is present in GTSeek data
    #If REF=C and ALT=T
    #The cells where the genotype is CC --> 0/0
    vcfFormat[a,which(geno[,tmp.col] == paste0(alleleState$REF[a],alleleState$REF[a]))] <- "0/0"
    #Cells where genotype is CT or TC --> 0/1
    vcfFormat[a,which(geno[,tmp.col] == paste0(alleleState$REF[a],alleleState$ALT[a]) |
                        geno[,tmp.col] == paste0(alleleState$ALT[a],alleleState$REF[a]))] <- "0/1"
    #Cells where genotype is TT --> 1/1
    vcfFormat[a,which(geno[,tmp.col]==paste0(alleleState$ALT[a],alleleState$ALT[a]))] <- "1/1"
  }
}
dim(vcfFormat)
```

```
## [1]  930 1024
```

```
#There will be lots of missing data here
vcfFormat[1:20,1:5]
```

```
##       [,1]  [,2]  [,3]  [,4]  [,5]
##  [1,] "./." "./." "./." "./." "./."
##  [2,] "./." "./." "./." "./." "./."
##  [3,] "./." "./." "./." "./." "./."
##  [4,] "./." "./." "./." "./." "./."
##  [5,] "1/1" "1/1" "0/1" "0/1" "1/1"
##  [6,] "./." "./." "./." "./." "./."
##  [7,] "./." "./." "./." "./." "./."
##  [8,] "./." "./." "./." "./." "./."
##  [9,] "./." "./." "./." "./." "./."
## [10,] "./." "0/1" "0/1" "0/1" "1/1"
## [11,] "./." "./." "./." "./." "./."
## [12,] "./." "./." "./." "./." "./."
## [13,] "0/0" "0/1" "0/0" "./." "0/0"
## [14,] "./." "./." "./." "./." "./."
## [15,] "1/1" "0/1" "0/0" "0/1" "0/0"
## [16,] "./." "./." "./." "./." "./."
## [17,] "./." "./." "./." "./." "./."
## [18,] "./." "./." "./." "./." "./."
## [19,] "./." "./." "./." "./." "./."
## [20,] "./." "./." "./." "./." "./."
```

```
#Do these match?
test.row <- which(apply(vcfFormat,1,FUN=function(x) length(unique(x))) > 1)[1]
alleleState[test.row,]
```

```
##            CHROM     POS REF ALT
## 5 NW_025576306.1 4455299   A   G
```

```
gt[1:5,paste0(alleleState$CHROM[test.row],"_",alleleState$POS[test.row])]
```

```
## [1] "GG" "GG" "GA" "GA" "GG"
```

```
vcfFormat[test.row,1:5]
```

```
## [1] "1/1" "1/1" "0/1" "0/1" "1/1"
```

```
#How many SNPs are in the dataset (i.e., not all ./.)
sum(apply(vcfFormat, 1, FUN=function(x) length(unique(x))) > 1)
```

```
## [1] 132
```

```
#Now write this to a file...
colnames(vcfFormat) <- paste0("GTS_",gt.smd$ugaID[-drop.me])
dim(vcfFormat)
```

```
## [1]  930 1024
```

```
write.table(vcfFormat, sep = "\t", file = "extraCols.vcf",
            quote = FALSE, row.names = FALSE, col.names = TRUE)
```

It seems like this worked. Now we append it to the VCF (`cbind` in R = `paste` in bash) and save the output. Let's try to do this by first condensing the vcf into a simpler format. Note that we're working with a simple VCF output by `PLINK` as it is. The "FORMAT" here is only GT (genotypes only, no depth, quality, allele depth, genotype likelihoods, etc.).

```
#Drop all header lines except the column labels
grep -v ^## LGC_allSCRsnps.vcf > lgc.vcf
#Paste (cbind) the SNP rows of the LGC_SCR with the extraCols object created above
paste lgc.vcf extraCols.vcf > out.vcf
#Now retrieve the header lines
grep ^## LGC_allSCRsnps.vcf > fullout.vcf
#And append the combined genotype object to the headers
cat out.vcf >> fullout.vcf
```

Check to make sure that all loci have exactly two alleles. Some of the markers from the GTSeek panel have 3!

```
vcftools --vcf fullout.vcf --out fullout \
         --freq
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf fullout.vcf
    --freq
    --out fullout

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
```

```
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
After filtering, kept 1851 out of 1851 Individuals
Outputting Frequency Statistics...
After filtering, kept 930 out of a possible 930 Sites
Run Time = 1.00 seconds
```

Now read this file into R and identify sites that have exactly two alleles in the dataset...

```
snpfreq <- read.table("fullout.frq", skip = 1, fill=TRUE)
colnames(snpfreq)[3] <- "N_ALLELES"
biallelic <- snpfreq[which(snpfreq$N_ALLELES == 2),c(1,2)]

fixed.sites <- c(grep(":1", biallelic$A1),
                 grep(":1", biallelic$A2),
                 grep(":1", biallelic$A3))
fixed.sites
```

```
## integer(0)
```

```
biallelic[fixed.sites,]
```

```
## [1] V1 V2
## <0 rows> (or 0-length row.names)
```

```
if(length(fixed.sites) > 0) {
  biallelic <- biallelic[-fixed.sites,c(1,2)]
}

nrow(biallelic)
```

```
## [1] 925
```

```
write.table(biallelic, file = "biallelic.txt",
            quote = FALSE, row.names = FALSE, col.names = FALSE)
```

Now keep only this list of SNP loci...

```
vcftools --vcf fullout.vcf --out prefinal \
        --positions biallelic.txt \
        --remove-filtered-all --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf fullout.vcf
    --out prefinal
    --positions biallelic.txt
    --recode
    --remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
After filtering, kept 1851 out of 1851 Individuals
Outputting VCF file...
After filtering, kept 925 out of a possible 930 Sites
Run Time = 0.00 seconds
```

This all worked, so we should be in good shape for downstream analyses with this file!

Individual-level filter (missingness)

Lastly, we want to check to see how many SNPs are genotyped for each individual in the dataset and combine our metadata files into a version with both LGC and GTSeek samples! We also want to remove samples that are genotyped at fewer than 100 SNPs (GTSeek) or <50% of the SNPs in the combined dataset (LGC).

```
#Get missing data information for each individual
system("vcftools --vcf prefinal.recode.vcf --out prefinal --missing-indv")

#Read this file into R
missSNPs <- read.table("prefinal.imiss", header = TRUE)

#Combine the two metadata files - LGC & GTSeek
```

```r
lgc.smd <- read.csv("../sampleinfo/sampleMetadata_nSNPgt.csv", header = TRUE)
colnames(lgc.smd)[1] <- "extID"
colnames(gt.smd)[1] <- "extID"
smd <- rbind(lgc.smd, gt.smd)

smd$vcfID <- c(lgc.smd$extID,
               paste0("GTS_",gt.smd$ugaID))
smd$FILT <- NA

#Update nSNPgt for all samples
for(i in 1:nrow(smd)) {
  tmp <- which(missSNPs$INDV == smd$vcfID[i])
  if(length(tmp) > 0) {
    smd$nSNPgt[i] <- missSNPs$N_DATA[tmp]-missSNPs$N_MISS[tmp]
  }
}

#Set the thresholds
GTS_thresh <- 100
LGC_thresh <- missSNPs$N_DATA[1]*0.5

for(i in 1:nrow(smd)) {
  tmp <- which(missSNPs$INDV == smd$vcfID[i])
  if(length(tmp) > 0) {
    if(length(grep("GTS_",smd$vcfID[i])) > 0) {
      if(smd$nSNPgt[i] < GTS_thresh) {
        smd$FILT[i] <- TRUE
      } else {
        smd$FILT[i] <- FALSE
      }
    } else if(length(grep("UGA_",smd$vcfID[i])) > 0) {
      if(smd$nSNPgt[i] < LGC_thresh) {
        smd$FILT[i] <- TRUE
      } else {
        smd$FILT[i] <- FALSE
      }
    }
  } else {
    smd$FILT[i] <- TRUE
  }
  rm(tmp)
}

#Number removed from dataset
sum(smd$FILT)
```

```
## [1] 552
```

```r
#Number NOT removed
sum(!smd$FILT)
```

```
## [1] 1653
```

```r
#Number of GTSeek samples retained
sum(smd$nSNPgt[grep("GTS_", smd$vcfID)] >= GTS_thresh, na.rm = TRUE)
```

```
## [1] 993
```

```r
#Number of LGC samples with 100+ SNPs genotyped
sum(smd$nSNPgt[grep("UGA_", smd$vcfID)] >= LGC_thresh, na.rm = TRUE)
```

```
## [1] 660
```

```r
#These should match
sum(!smd$FILT) == sum(smd$nSNPgt[grep("GTS_", smd$vcfID)] >= GTS_thresh, na.rm = TRUE) +
  sum(smd$nSNPgt[grep("UGA_", smd$vcfID)] >= LGC_thresh, na.rm = TRUE)
```

```
## [1] TRUE
```

```r
#Number of SNPs genotyped per sample
smd$nSNPgt[which(smd$FILT == FALSE)]
```

```
##    [1] 835 882 745 876 677 821 660 870 878 893 837 913 557 862 754 913 921 552
##   [19] 880 922 911 917 865 901 794 819 916 881 875 867 917 914 916 916 880 635
##   [37] 859 756 688 892 920 917 915 892 910 919 762 855 908 918 909 875 894 882
##   [55] 917 914 912 800 904 907 910 913 914 886 665 903 792 912 887 754 911 919
##   [73] 913 882 901 920 917 907 911 922 917 904 691 901 785 845 881 865 912 872
##   [91] 919 904 916 916 907 915 719 913 914 907 850 904 802 911 901 845 848 902
##  [109] 699 888 741 795 853 806 889 774 864 808 876 680 914 917 898 888 731 917
##  [127] 871 910 906 911 773 908 728 912 918 899 873 884 888 898 621 911 912 672
##  [145] 915 918 918 885 915 687 918 917 869 903 916 910 782 870 911 911 900 858
##  [163] 916 917 918 901 922 873 745 921 917 918 889 923 889 919 917 918 915 920
##  [181] 916 898 920 747 693 919 917 920 917 920 917 915 894 765 906 913 566 732
##  [199] 918 904 799 911 884 826 662 897 914 911 889 917 829 913 920 745 866 917
##  [217] 569 792 680 922 922 921 917 853 920 525 919 700 658 903 922 917 774 918
##  [235] 903 916 904 647 900 913 848 919 911 918 919 911 759 491 917 909 475 896
##  [253] 723 917 791 914 546 473 599 914 912 917 897 918 911 922 922 538 916 913
##  [271] 918 910 919 752 887 920 919 915 913 908 920 917 920 922 884 919 910 915
##  [289] 899 886 900 913 724 920 917 918 883 837 909 919 921 919 919 761 677 921
##  [307] 911 921 847 922 922 828 918 831 922 913 919 919 921 921 919 607 915 913
##  [325] 921 918 685 922 858 914 921 914 922 921 916 838 913 754 906 914 646 849
##  [343] 921 921 920 912 920 902 913 920 915 808 921 922 918 922 913 914 885 901
##  [361] 836 888 823 902 887 915 912 877 919 915 569 908 913 902 582 836 913 885
##  [379] 892 904 852 877 889 550 897 909 563 913 919 916 887 915 839 915 917 912
##  [397] 917 919 911 915 911 918 920 921 858 735 829 875 918 692 905 917 918 917
##  [415] 896 914 914 913 845 917 863 903 907 899 878 880 914 916 916 761 885 913
##  [433] 905 918 880 669 892 903 910 886 910 608 907 913 909 906 917 918 873 918
##  [451] 916 918 917 654 914 763 918 575 812 897 913 915 907 848 793 896 567 647
##  [469] 915 828 919 904 767 897 895 562 902 815 916 915 854 632 891 743 653 855
##  [487] 900 905 666 917 471 911 916 873 844 915 784 855 897 885 893 525 766 896
##  [505] 828 917 888 853 463 912 621 913 905 469 906 729 905 888 842 874 915 913
##  [523] 794 599 909 896 915 698 900 910 787 867 499 510 508 842 906 828 868 899
##  [541] 894 708 865 850 745 819 604 909 750 726 911 917 904 913 801 915 699 916
##  [559] 917 907 874 915 910 906 797 468 896 884 887 902 692 862 913 862 656 901
```

```
##  [577] 643 918 725 576 914 912 898 896 866 896 833 915 827 909 680 886 916 907
##  [595] 914 915 586 861 537 910 550 915 901 916 900 914 822 776 879 910 903 903
##  [613] 901 882 902 900 879 858 908 895 897 882 909 910 491 905 911 810 911 906
##  [631] 908 882 894 844 844 913 564 908 586 746 899 903 578 910 885 863 903 609
##  [649] 897 908 905 722 861 904 892 504 761 518 914 913 104 122 121 123 123 118
##  [667] 121 123 122 117 122 121 123 118 123 122 121 123 122 126 122 122 113 123
##  [685] 120 124 122 121 122 124 124 120 122 116 123 118 119 120 123 123 123 120
##  [703] 123 123 125 120 122 120 123 120 123 125 121 123 122 122 119 123 123 123
##  [721] 121 122 119 124 118 123 122 124 123 115 121 120 124 121 122 114 122 123
##  [739] 121 120 121 117 125 119 123 119 117 119 123 114 124 120 124 122 121 122
##  [757] 122 123 119 124 122 122 116 121 121 122 119 123 114 122 123 122 123 122
##  [775] 119 125 123 124 122 123 123 112 125 123 122 121 121 123 119 117 123 121
##  [793] 123 119 123 123 120 122 120 123 122 121 122 122 120 117 125 121 118 122
##  [811] 122 121 119 121 120 123 122 116 119 120 123 123 122 124 122 125 120 125
##  [829] 120 117 123 121 125 122 124 123 124 124 119 121 121 123 121 122 124 121
##  [847] 124 118 121 121 124 124 124 123 123 122 123 113 120 117 124 122 124 123
##  [865] 124 121 118 124 123 123 123 122 124 122 123 119 124 123 124 123 122 120
##  [883] 122 120 118 122 126 121 123 121 122 122 118 124 120 123 121 124 120 124
##  [901] 122 123 123 123 122 118 119 123 121 125 121 122 122 126 121 120 122 118
##  [919] 121 122 123 122 125 123 124 122 121 124 122 119 124 122 119 126 121 124
##  [937] 120 123 120 123 122 120 119 122 122 123 123 124 122 122 124 121 123 118
##  [955] 124 119 115 122 123 122 122 120 121 123 120 121 120 125 123 120 123 124
##  [973] 124 122 124 125 125 122 123 119 122 121 123 122 120 121 123 120 124 123
##  [991] 122 117 124 122 123 126 124 123 122 126 124 125 124 123 124 117 121 123
## [1009] 122 122 124 124 124 123 123 122 123 123 124 122 125 123 125 124 123 122
## [1027] 124 123 123 121 122 122 123 123 123 122 124 124 121 123 123 115 123 124
## [1045] 120 123 116 125 121 125 123 124 122 122 124 121 123 125 121 124 122 123
## [1063] 121 123 124 124 122 123 124 123 122 120 121 125 123 124 126 123 122 124
## [1081] 123 122 122 120 123 124 123 123 124 123 122 123 122 121 123 122 124 123
## [1099] 124 122 124 123 123 125 125 124 123 122 123 124 112 122 123 123 123 126
## [1117] 123 122 123 123 124 123 123 123 124 125 118 117 124 123 121 123 124 125
## [1135] 124 123 123 124 122 123 123 124 119 126 124 124 122 124 124 116 124 125
## [1153] 122 123 123 118 124 124 123 125 124 124 123 124 122 124 123 121 123 123
## [1171] 125 122 125 123 124 122 125 123 125 125 123 122 120 122 125 124 122 125
## [1189] 124 122 121 123 119 122 125 124 123 122 124 123 125 125 122 125 125 122
## [1207] 124 125 120 123 124 122 124 123 120 124 124 121 121 120 124 123 124 124
## [1225] 124 123 124 123 126 123 122 120 123 121 124 125 122 122 121 121 123 120
## [1243] 121 124 125 122 125 125 126 124 122 123 123 123 123 124 123 120 124 124
## [1261] 123 122 123 122 121 122 123 124 122 121 123 122 123 124 122 121 124 123
## [1279] 123 123 124 124 124 125 123 120 123 121 123 120 123 123 122 123 121 119
## [1297] 123 121 124 124 124 124 124 125 124 124 121 124 119 123 123 122 123 125
## [1315] 122 123 125 126 125 125 123 126 126 124 125 124 123 120 123 122 121 123
## [1333] 121 124 123 123 124 122 124 123 124 124 124 124 116 123 123 122 123 124
## [1351] 123 124 123 124 122 123 123 119 121 124 122 124 124 122 123 123 124 124
## [1369] 124 122 121 122 124 125 124 122 122 123 123 122 122 124 123 124 125 124
## [1387] 123 122 124 125 124 122 124 124 124 122 123 122 122 124 126 120 125 124
## [1405] 124 123 122 121 120 123 123 125 125 123 124 122 122 124 123 121 120 122
## [1423] 122 124 122 124 123 124 124 120 123 120 121 121 124 124 124 120 124 121
## [1441] 123 125 124 119 121 124 122 116 124 122 125 126 123 124 123 123 123 115
## [1459] 124 122 123 123 122 120 124 124 119 121 121 123 124 124 124 124 123 123
## [1477] 124 124 124 119 122 119 122 124 120 123 124 120 121 104 124 119 124 109
## [1495] 123 124 124 122 123 124 119 124 118 124 122 122 124 123 113 125 125 123
## [1513] 124 123 123 123 121 122 123 114 123 125 124 122 123 124 123 124 124 125
## [1531] 115 112 120 122 123 117 120 124 124 110 121 122 121 123 122 120 123 123
```

```
## [1549] 120 123 125 125 119 123 118 116 125 124 113 124 123 123 125 123 124 119
## [1567] 123 119 124 123 116 123 116 124 124 123 123 124 125 123 120 120 122 124
## [1585] 124 124 124 123 123 123 113 123 112 123 122 110 120 122 124 116 123 123
## [1603] 124 119 122 113 109 122 124 120 124 123 124 120 122 122 115 123 120 121
## [1621] 124 125 123 123 124 122 123 123 122 119 122 123 124 125 118 122 116 125
## [1639] 121 123 123 123 118 121 117 117 124 124 122 123 124 123 120
```

```
#Write metadata file
write.csv(smd, "sampleMetadata_combined.csv", quote = FALSE, row.names=FALSE)

#Keep individuals that are not filtered from the dataset
keeplist <- smd$vcfID[which(smd$FILT == FALSE)]

#Write keeplist to a text file
write.table(keeplist, file = "keeperInds.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
```

Now apply the individual-level filter...

```
vcftools --vcf prefinal.recode.vcf --out finalSCR \
         --keep keeperInds.txt --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf prefinal.recode.vcf
    --keep keeperInds.txt
    --out finalSCR
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Keeping individuals in 'keep' list
After filtering, kept 1653 out of 1851 Individuals
```

```
Outputting VCF file...
After filtering, kept 925 out of a possible 925 Sites
Run Time = 0.00 seconds
```

Evaluate Power of the Set

We can use the allele frequencies above to calculate p_{ID} for the marker set (and for random subsets of the SNP data to give us an idea of how many markers we need to successfully genotype to keep an individual in the dataset).

Here we're using the simple equation from Waits et al. (2001).
For one locus with two alleles:

$$p_{\mathrm{ID}} = \sum(p_i^4) + (2p_i p_j)^2$$

Then the power of the combined marker set is the product of p_{ID} across all SNP loci, as markers are assumed to provide independent information on identity.

First, extract allele frequencies for all loci in the combined dataset.

```
#Extract allele frequencies for these SNPs to calculate p_{ID}
plink2 --vcf finalSCR.recode.vcf --allow-extra-chr --freq --out finalSCR --silent
```

Now use these to calculate p_{ID}.

```
freqs <- read.table("finalSCR.afreq", header = FALSE)
colnames(freqs) <- c("CHROM", "CPOS", "REF", "ALT", "FRQ", "NCHR")
freqs$pID <- rep(NA, nrow(freqs))
for(l in 1:nrow(freqs)) {
  freqs$pID[l] <- (freqs$FRQ[l]^4+(1-freqs$FRQ[l])^4) +
                    (2*freqs$FRQ[l]*(1-freqs$FRQ[l]))^2
}
sum(log10(freqs$pID))
```

```
## [1] -338.1137
```

A more conservative estimate of the power of the set is provided by p_{IDsib}, which measures the same probability for siblings. Also from Waits et al. (2001)...

$$p_{\mathrm{IDsib}} = 0.25 + \left(0.5\sum p_i^2\right) + \left[0.5\left(\sum p_i^2\right)^2\right] - \left(0.25\sum p_i^4\right)$$

As above, the p_{IDsib} for the dataset is the product across loci.

```
freqs$pIDsib <- rep(NA, nrow(freqs))
for(l in 1:nrow(freqs)) {
  freqs$pIDsib[l] <- 0.25 + (0.5*sum(c(freqs$FRQ[l]^2,(1-freqs$FRQ[l])^2))) +
                      (0.5*(sum(c(freqs$FRQ[l]^2, (1-freqs$FRQ[l])^2)))^2) -
                      (0.25*sum(c(freqs$FRQ[l]^4, (1-freqs$FRQ[l])^4)))
}
sum(log10(freqs$pIDsib))
```

```
## [1] -175.141
```

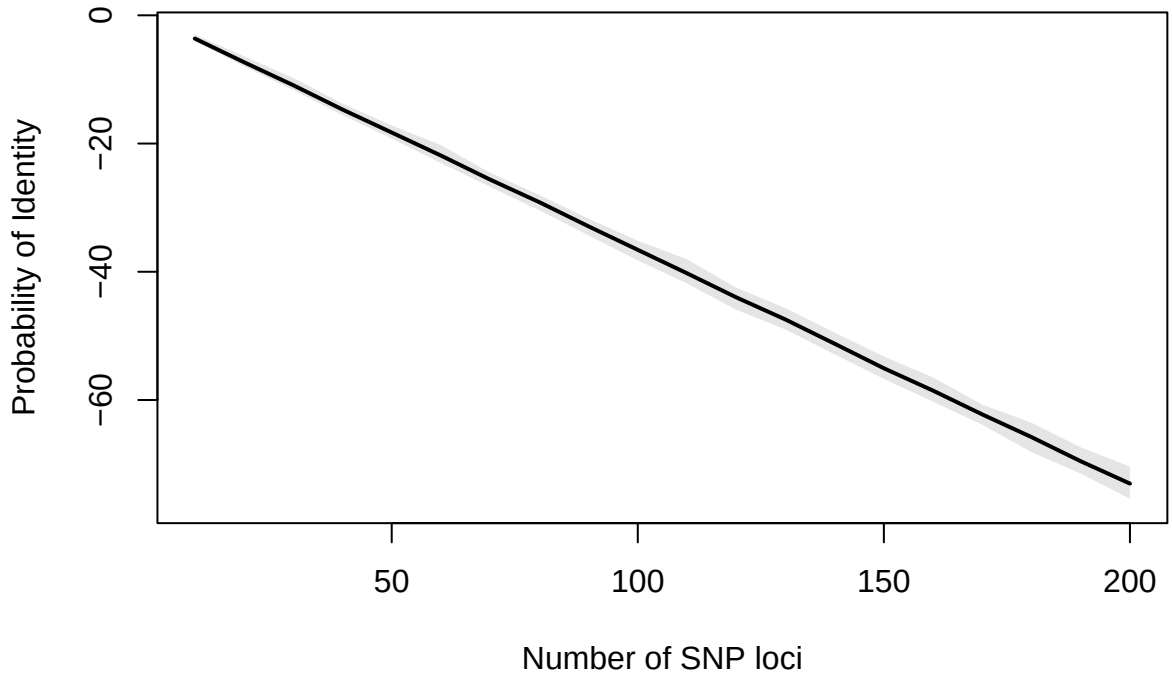
Power with subsets of the panel?

We have exceptional power for individual identification with the full marker set. However, few if any samples will be genotyped at all loci. Here we recalculate the probabilities above with random samples from the full panel (between 10 and 200 SNPs).

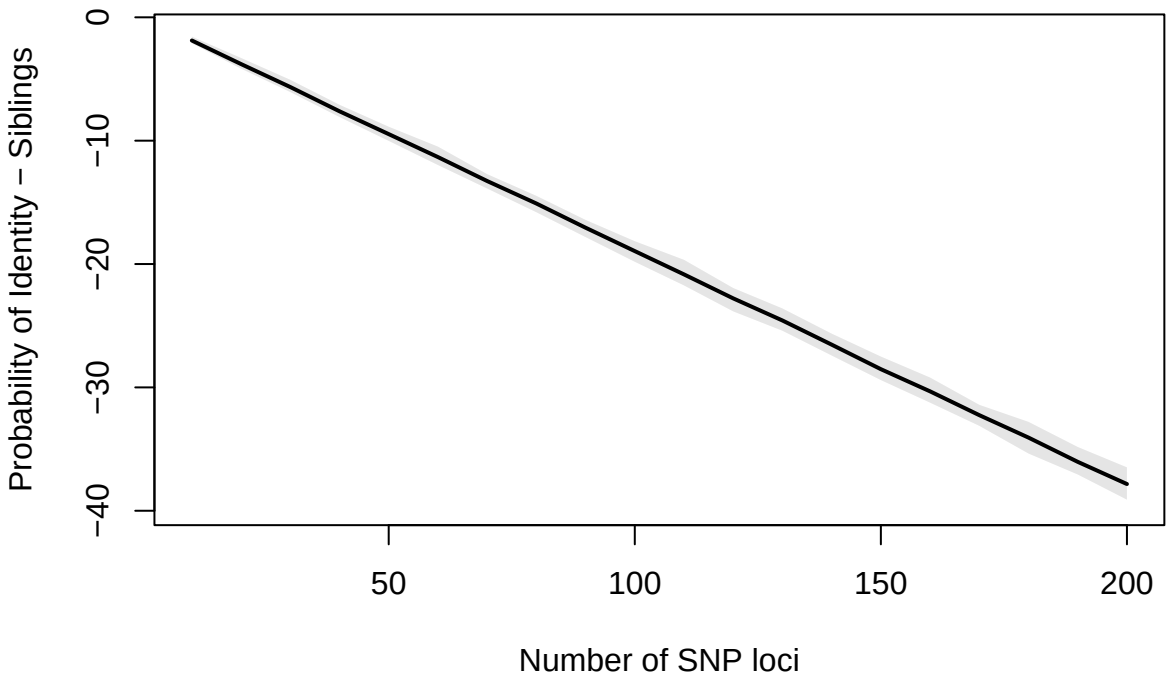
```
subsize <- seq(from = 10, to = 200, by = 10)
pIDsub <- matrix(data = NA, nrow = 100, ncol = length(subsize))
pIDsibsub <- matrix(data = NA, nrow = 100, ncol = length(subsize))

for(size in 1:length(subsize)) {
  for(rep in 1:100) {
    tmp.set <- sample(c(1:nrow(freqs)), subsize[size], replace = FALSE)
    tmp.freq <- freqs[tmp.set,]
    tmp.pID <- c()
    tmp.pIDsib <- c()
    for(l in 1:subsize[size]) {
      tmp.pID[l] <- (tmp.freq$FRQ[l]^4+(1-tmp.freq$FRQ[l])^4) +
                      (2*tmp.freq$FRQ[l]*(1-tmp.freq$FRQ[l]))^2
      pIDsub[rep,size] <- sum(log10(tmp.pID))
      tmp.pIDsib[l] <-  0.25 +
                          (0.5*sum(c(tmp.freq$FRQ[l]^2,(1-tmp.freq$FRQ[l])^2))) +
                          (0.5*(sum(c(tmp.freq$FRQ[l]^2, (1-tmp.freq$FRQ[l])^2)))^2) -
                          (0.25*sum(c(tmp.freq$FRQ[l]^4, (1-tmp.freq$FRQ[l])^4)))
      pIDsibsub[rep,size] <- sum(log10(tmp.pIDsib))
    }
    rm(tmp.pID, tmp.pIDsib)
    rm(tmp.set, tmp.freq)
  }
}


#Plot the results
library(scales)
plot(subsize, colMeans(pIDsub), type = "l",
     ylim = range(pIDsub), lwd = 2, col = "black",
     xlab = "Number of SNP loci", ylab = "Probability of Identity")
quant95 <- apply(pIDsub, 2, quantile, prob = c(0.025, 0.975))
polygon(x = c(subsize, subsize[length(subsize):1]),
        y=c(quant95[1,], quant95[2,][ncol(quant95):1]),
        col=alpha("black", 0.1), border = NA)
```




```
plot(subsize, colMeans(pIDsibsub), type = "l",
     ylim = range(pIDsibsub), lwd = 2, col = "black",
     xlab = "Number of SNP loci", ylab = "Probability of Identity - Siblings")
quant95 <- apply(pIDsibsub, 2, quantile, prob = c(0.025, 0.975))
polygon(x = c(subsize, subsize[length(subsize):1]),
        y=c(quant95[1,], quant95[2,][ncol(quant95):1]),
        col=alpha("black", 0.1), border = NA)
```



Test file

As a final test, we'll attempt to use this file with `PLINK` and `vcftools`...

```
#Attempt to calculate pairwise kinship coefficients in plink2
plink2 --vcf finalSCR.recode.vcf --make-king-table \
       --out testFinalSCR --allow-extra-chr --silent

#Attempt to use vcftools to make a 012 format file
vcftools --vcf finalSCR.recode.vcf --012 --out testFinalSCR

#Cleanup
rm LGC_allSCRsnps.vcf extraCols.vcf lgc.vcf out.vcf
rm testFinalSCR.log testFinalSCR.kin0
rm *012*
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf finalSCR.recode.vcf
    --012
    --out testFinalSCR
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
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Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
After filtering, kept 1653 out of 1653 Individuals
Writing 012 matrix files ... Done.
After filtering, kept 925 out of a possible 925 Sites
Run Time = 0.00 seconds
```

Proceed to recapture analyses in `PLINK` and `COLONY`!
